Component Identification List

PART NUMBER	QTY PER	DESCRIPTION	MANUFACTURER & TYPE NO.	
00022022	1	(41-02) Plastic Case	Airmold	RA21/16SC Black
00024074	1	(10-07) Label for Micromodule 85		
00024084	1	MM 85 Information Label		
00042180	1	(02-65) PC Board for Micromodule 85	Tingstal Pentaflex	F10937PC
00060017	1	(06-58) 34-Contact Cable Header	3M	3494-2003
00064077	7	(06-53) 14-Pin IC Socket	Burndy	DILB14P11
00064095	3	(06-55) 16-Pin IC Socket	Burndy	DILB16P11
00064102	2	(06-56) 18-Pin IC Socket	Burndy	DILB18P11
00064111	3	(06-57) 40-Pin IC Socket	Burndy	DILB40P11
00067190	1	(06-31) Red Banana Jack	E.F.Johnson	108-09-02-001
00067207	1	(06-30) Black Banana Jack	E.F.Johnson	108-09-03-001
00070524	2	(32-209) #6 × 1/2" Sheet Metal Screw	Abbott	Cad. Type A
00078026	2	(16-44) Mounting Clip No. 6	Anchor Bolt	C13090-6A
00079034	2	(40-.02) 6" × 5/8" Loop Fastener	3M	No.3542
00213807	1	(24-75122) 22 k ohm, 1/4W, 5% Resistor		
00251089	1	(25-100) 22 ohm Resistor Network	TRW CTS	516-3-220G 761-3-R22
00251605	1	(25-101) 3.3 k ohm Resistor Network	TRW CTS	516-3-332G 761-3-R3.3K
00290153	1	(04-45) One-Button Module Switch	Grayhill	87YY3201A-538
00295005	1	(04-46) Twelve-Button Alpha-Numeric Keypad	Grayhill	87YY3612A-505
00295014	1	(04-48) Twelve-Button Alpha Keypad	Grayhill	87YY3612A-506
00301800	1	(28-22006) 20 pF Disc Capacitor	RMC	20% NPO
00309259	10	(30-450001) .1 MFD Monolythic Capacitor	Erie AVX	3439-050-651-104 SR21E104MAA
00400051	1	(34-T159) 1N4004 Diode		
00430019	2	(45-10)74LS138 3-to-8 Decoders	Japan Signetics TI	M74LS138N N74LS138N SN74LS138N

PART NUMBER	QTY PER	DESCRIPTION	MANUFACTURER & TYPE NO.	
00430028	1	(45-11) 74LS145 BCD-to-Decimal Decoder Drivers	Japan Signetics TI	M77LS145P N74LS145N SN74LS145N
00430037	1	(45-12) 74LS10 Triple 3-Input NAND Gate	Japan Signetics TI	M74LS10D N74LS10N SN74LS10N
00440106	1	(45-09) UDN6118A Source Drivers	EXAR Sprague	6118P UDN6118A
00450006	1	(45-07) 8185 1K × 8-Bit RAM	Intel	P8185
00460004	1	(45-08) 8085 Microprocessor	Toshiba Intel Intel	TMP8085P P8085A P8085AH
00460013	1	(45-05) 8155 RAM, I/O and Timer	Toshiba Intel Intel	TMP8155P P8155P P8155PH
00460022	1	(45-06) 8755 EPROM and I/O	Toshiba Intel NEC	TMP8755AC D8755A D8755A
00480000	6	(34-47) LED Display MAN 74A	Gen.Inst.	MAN74A 028D
00560021	1	(46-02) Crystals 6.144 MHz	CTS	MP-061

BUSS CONNECTER PIN-OUT

Pin #	Description
1	HOLD
2	HOLD A
3	CLOCK
4	SID
5	TRAP
6	RST 7.5
7	$\overline{RD}$
8	INTR
9	$\overline{INTR\ A}$
10	A15
11	A14
12	A13
13	A12
14	A11
15	A10
16	A9
17	A8

Pin #	Description
18	VSS
19	AD7
20	AD6
21	AD5
22	AD4
23	AD3
24	AD2
25	AD1
26	AD0
27	ALE
28	$\overline{WR}$
29	RST 6.5
30	IO/$\overline{M}$
31	RDY
32	SOD
33	RESET OUT
34	VCC

User Accessible Subroutine

HLFORM-012F Hex -

This subroutine will format a binary character in Register A to a display character and then store it in the Display Buffer.

Enter subroutine with character to be displayed in Register A and storage location HLDISP pointing to Display Buffer.

Registers A, H, L & Flag will be destroyed.

DSDATA-0144 Hex -

The data pointed by H & L Registers is formated by this subroutine and stored in the Display Buffer.

Enter subroutine with Registers H & L pointing to the data to be converted to a display character and storage location HLDISP pointing to Display Buffer.

Registers A, H, L & Flag will be destroyed.

SCAN-015B Hex -

This subroutine will scan, debounce and decode keypad. In addition it will display characters and refresh the LED display.

Enter subroutine with display characters in Display Buffer. The subroutine will then display the characters on the LED's. It will return from subroutine when any key is pressed. When returning, the binary representation of the key pressed is in Register A.

All registers are destroyed.

INPUT4-01C9 Hex -

This subroutine will input, display and return with four characters in Registers H & L. Four characters typed in on the keypad are converted to display characters and displayed on the LED's. Also, inputted characters are converted to binary and on return from this subroutine the binary is in the H & L Registers.

On entry to subroutine no set-up is needed.

All registers are destroyed.

INPUT2-01E0 Hex -

This subroutine will input, display and return with two characters in Register A. Two characters typed in on the keypad are converted to a display character and displayed on the LED's. Also, inputted characters are converted to binary, and on return from this subroutine, the binary is in Register A.

On entry to subroutine no set-up is needed.

All registers are destroyed.

INFROM-022A Hex -

The data in the input buffer (INBUFF) will be converted from input characters to binary. This subroutine will convert either two (2) or four (4) characters, with character count stored in DGTCNT.

Enter subroutine with characters to be converted in the INBUFF and character count in DGTCNT.

Subroutine will convert two input characters into one byte in Register A, or four input characters into two bytes in Registers H & L.

Registers A, B, H, L & Flag will be destroyed.

CONHEX-024D Hex -

This subroutine provides a search function of a three-dimensional table.

Enter this subroutine with binary characters in Register A and it will return with display characters in Register A and in Display Buffer.

Storage location HLDISP must contain current location of Display Buffer.

Registers A, H, L & Flag will be destroyed.

DISREG-02E0 Hex -

The CPU registers will be displayed. Upon entry, the Address portion of the LED display will show the Program Counter contents. On each depression of the "X" command key another register pair will be displayed. Each register pair is indicated by a two-letter designation in the Data portion of the LED display.

MEMTEST-0500 Hex -

Test routine to test 1K RAM at location 1400 Hex.

No set-up required on entry.

All registers are destroyed.

If error is found location is displayed by entry into DISREG subroutine. P.C. will contain location, Register A contains data from RAM and Register B contains data that RAM failed. on.

On end of memory test the LED test will be executed. Also see Step 6 of test procedures.

LEDTEST-0530 Hex -

This routine will test the student I/O ports by turning on every other LED (see Step 6 of test procedures).

No set-up required on entry.

Only Register A is displayed.

DISPLAY-0560 Hex -

This subroutine will display the contents of Register A in the data portion of the LED display for approximately one second. The display is cleared on entry to and return from this subroutine.

On entry data to be displayed is in Register A.

All registers are saved and restored by this subroutine.

GET-05A0 Hex -

When this subroutine is called, a decimal point is displayed on the left-most LED digit. This indicates that any key may be pressed and value of the key is returned in Register A.

No set-up is required.

Registers A and Flag are destroyed.

CLEAR-05B2 Hex -

This routine will clear (blank) the LED display.

No set-up is required.

No registers are destroyed.

FIVEØØ-05C1 Hex -

This subroutine is a 500 microsecond delay.

No set-up is required.

No registers are destroyed.

BEEP-05D0 Hex -

This subroutine will produce a 1 kHz tone on the S.O.D. pin of the student I/O (DIP-B, Pin 14) for approximately half a second.

No set-up is required.

No registers are destroyed.

```
    ASM V7-15MON.BZX
CP/M ASSEMBLER - VER 2.0


                ;
                ;8085 MONITOR V7.15
                ;COPYRIGHT (C) 1982
                ;BELL & HOWELL EDUCATION GROUP
                ;AUTHOR RICHARD C. AUBERT
                ;DATE 03/08/83
                ;
                ;MONITOR I/O
                ;
0000 =          DRA:    EQU     00H     ;8755 DATA PORT A
0001 =          DRB:    EQU     01H     ;8755 DATA PORT B
0002 =          DDRA:   EQU     02H     ;8755 DATA DIRECTION PORT A
0003 =          DDRB:   EQU     03H     ;8755 DATA DIRECTION PORT B
0008 =          CSR1:   EQU     08H     ;8155 DATA COMMAND / STATUS REG.
000C =          TIMERL: EQU     0CH     ;8155 TIMER PORT LOW BYTE
000D =          TIMERH: EQU     0DH     ;8155 TIMER PORT HIGH BYTE
                ;
                ;MONITOR RAM LOCATIONS
                ;
0000 =          BEGIN:  EQU     0000H   ;START OF MONITOR
08FF =          USRCSR: EQU     08FFH   ;USER CSR IMAGE
08FE =          DIGIT:  EQU     08FEH   ;KB DECODE CHAR.STORAGE
08FD =          DSPBFA: EQU     08FDH   ;DISPLAY ADDRESS BUFFER
08F8 =          DSPBFD: EQU     08F8H   ;DISPLAY DATA BUFFER
08F3 =          INBUFF: EQU     08F3H   ;INPUT BUFFER
08F0 =          HSAVE1: EQU     08F0H   ;SAVE AREA 1 FOR H&L
08EE =          HSAVE2: EQU     08EEH   ;SAVE AREA 2 FOR H&L
08EC =          HLDISP: EQU     08ECH   ;DISPLAY POINTER SAVE AREA
08EB =          DGTCNT: EQU     08EBH   ;SAVE AREA FOR COUNTING DIGITS
08EA =          DIGIT2: EQU     08EAH   ;KB DECODE CHAR. STORAGE AREA
08E8 =          STKPT:  EQU     08E8H   ;POINT TO TOP OF STACK IN S/S
08E6 =          BKSAVE: EQU     08E6H   ;POINT TO BOTTOM USER STACK
08E2 =          NOBKRG: EQU     08E2H   ;NO. OF BREAK POINTS & BUFFER
08E1 =          INTERF: EQU     08E1H   ;S/S INTERRUPT FLAG
08C0 =          STACK:  EQU     08C0H   ;START OF STACK
                ;
                ;USER INTERRUPT VECTORS         PUT JUMP ADDRESS HERE
                ;                               FOR INTERRUP VECTORS
                ;
08DC =          USER5:  EQU     08DCH   ;       RST 7.5
08D9 =          USER4:  EQU     08D9H   ;       RST 6.5
08D6 =          USER3:  EQU     08D6H   ;       RST 6
08D3 =          USER2:  EQU     08D3H   ;       RST 5.5 NOT SUPPORTED
                ;                               IN HARDWARE.
08D0 =          USER1:  EQU     08D0H   ;       RST 5
                ;
                ;USER DEFINED VECTORS
                ;
08CC =          USERP:  EQU     08CCH   ;INITIALIZED IN
08C9 =          USERL:  EQU     08C9H   ;START OF MONTOR
08C6 =          USERR:  EQU     08C6H   ;TO A JUMP TO COMMD
                ;
                ;MONITOR INTERRUPT VECTORS
                ;
00CF =          RST1:   EQU     0CFH    ;USED FOR BREAK POINT
08C3 =          UTIMER: EQU     08C3H   ;INIT. TO A JUMP TO DISREG
                ;
                ;*****************************************************
                ;* IF ANY CHANGES ARE MADE BEFORE 'DISREG' ROUTINE,  *
                ;* 2ND. BYTE IN 'UMOVE' (038BH) MUST BE CHANGED TO   *
                ;* REFLECT NEW LOCATION OF 'DISREG' ROUTINE.         *
                ;*****************************************************
                ;
                ;RST-7 USED TO RETURN TO COMMAND MODE(WARM START)
                ;RST-0 USED TO RETURN TO MONITOR (COLD START)
                ;TRAP USED FOR S/S OR BY USER.(INSERT JUMP
                ;                INSTRUCTION AT UTIMER)
                ;
                ;COMMAND KEY DECODE HEX VALUE
                ;
0010 =          X:      EQU     10H     ;EXCUTE
0011 =          P:      EQU     11H     ;USER DEFINED
0012 =          L1:     EQU     12H     ;USER DEFINED
0017 =          R:      EQU     17H     ;USER DEFINED
0013 =          W:      EQU     13H     ;WRITE / READ
0014 =          B1:     EQU     14H     ;BREAK POINT
0015 =          S:      EQU     15H     ;SINGLE STEP
```

```
0016 =          D1:     EQU     16H     ;DISPLAY CPU REG.
                ;
                ; MONITOR STARTS AT LOCATION 0 HEX
                ;
0000                    ORG     BEGIN
0000 31C008     START:  LXI     SP,STACK
0003 C37300             JMP     INITIZ
0006 00                 NOP
0007 00                 NOP
0008 F5         INTB:   PUSH    PSW     ;SAVE ALL CPU REG.
0009 C5                 PUSH    B       ;ON INTERRUPT FROM
000A D5                 PUSH    D       ;BREAK POINT.
000B E5                 PUSH    H
000C 21FCFF             LXI     H,0FFFCH        ;SAVE STACK POINTER
000F 39                 DAD     SP      ;POINTING TO WHAT IT
0010 E5                 PUSH    H       ;WOULD BE AT END OF STACK
0011 EB                 XCHG
0012 20                 DB      20H     ;"RIM INSTRUCTION"
0013 F5                 PUSH    PSW
0014 210E00             LXI     H,000EH ;POINT H&L TO PROGRAM COUNTER
0017 39                 DAD     SP      ;IN STACK
0018 CD4000             CALL    MODPC
                ;RESTORE ALL CPU REG. ON RETURN FROM INTERRUPT
                ;       (BREAK POINT & SINGLE STEP)
001B F1         RESREG: POP     PSW
001C 30                 DB      30H     ;"SIM INSTRUCTION"
001D E1                 POP     H       ;DUMMY POP FOR STACK POINTER
001E E1                 POP     H       ;PLACED ON STACK
001F D1                 POP     D
0020 C1                 POP     B
0021 F1                 POP     PSW
0022 00                 NOP
0023 C9                 RET
0024 C3C308     INTSS:  JMP     UTIMER  ;VECTORED JUMP ON S/STEP
                ;                       INTERRUPT
0027 00                 NOP
0028 C3D008     RST5:   JMP     USER1   ;FOR ALL USER OPERANDS
002B 00                 NOP             ;SEE USER INTERRUPT VECTORS
002C C3D308     RST55:  JMP     USER2
002F 00                 NOP
0030 C3D608     RST6:   JMP     USER3
0033 00                 NOP
0034 C3D908     RST65:  JMP     USER4
0037 00                 NOP
0038 C38A00     COM:    JMP     COMMD   ;WARM START ON RST-7
003B 00                 NOP
003C C3DC08     RST75:  JMP     USER5
003F 00                 NOP
                ;
                ; BREAK POINT ROUTINE
                ;
0040 EB         MODPC:  XCHG
0041 22E608             SHLD    BKSAVE  ;BACK-UP USER'S PC REG.
0044 EB                 XCHG            ;STORED IN STACK
0045 22E808             SHLD    STKPT
0048 2B                 DCX     H
0049 56                 MOV     D,M
004A 2B                 DCX     H
004B 5E                 MOV     E,M
004C 1B                 DCX     D
004D 73                 MOV     M,E
004E 23                 INX     H
004F 72                 MOV     M,D
0050 2B                 DCX     H
0051 22EE08             SHLD    HSAVE2
0054 21E208             LXI     H,NOBKRG
0057 7E                 MOV     A,M     ;DECREMENT NUMBER IN
0058 47                 MOV     B,A     ;BREAK POINTER COUNTER
0059 3D                 DCR     A
005A 77                 MOV     M,A     ;RETURN USER'S INSTRUCTION
005B 23                 INX     H       ;STORED IN BUFFER TO USER'S
005C 7E                 MOV     A,M     ;PROGRAM
005D EB                 XCHG
005E 77                 MOV     M,A
005F EB                 XCHG
0060 11E308             LXI     D,NOBKRG+1
0063 23         STEP:   INX     H       ;MOVE STORED USER'S
0064 7E                 MOV     A,M     ;INSTRUCTION
0065 EB                 XCHG            ;TO TOP OF BUFFER
0066 77                 MOV     M,A
0067 23                 INX     H
0068 05                 DCR     B
0069 C26300             JNZ     STEP
```

```
006C 2AE608            LHLD    BKSAVE  ;LOAD H&L WITH BOTTOM OF
006F EB                XCHG            ;STOCK AND MOVE IT TO D&E
0070 C30A03            JMP     DREG    ;GO DISPLAY CPU REG.
                       ;
                       ;INITIALIZATION ROUTINE (COLD START)
                       ;
0073 3EFF       INITIZ: MVI     A,0FFH  ;INITIALIZE 8755 I/O CHIP
0075 D303              OUT     DDRB    ;PORT B - BITS 4-7 ARE INPUTS
0077 3E07              MVI     A,07H   ;PORT A & PORT B - BITS 0-2
0079 D302              OUT     DDRA    ;ARE OUTPUTS
007B 060C              MVI     B,0CH
007D 11C308            LXI     D,UTIMER;MOVE RST 7 (FF) TO USER
0080 218B03            LXI     H,UMOVE ;DEFINED SUBROUTINES
0083 CDE700            CALL    MOVE
0086 AF                XRA     A       ;MOVE ZERO TO # OF BREAK POINTS
0087 32E208            STA     NOBKRG
                       ;
                       ;COMMAND ROUTINE
                       ;
008A 31C008     COMMD:  LXI     SP,STACK;OUTPUTS PROMPT INDICATION
008D AF                XRA     A       ;AND SCANS KEYPAD FOR
008E 32E108            STA     INTERF  ;COMMAND KEY
0091 CDDF00            CALL    MOVE1
0094 CD5B01            CALL    SCAN
0097 FE13              CPI     W
0099 CAF200            JZ      WRITE
009C FE15              CPI     S
009E CA5903            JZ      SINSTP
00A1 FE16              CPI     D1
00A3 CCE002            CZ      DISREG
00A6 CA8A00            JZ      COMMD
00A9 FE10              CPI     X
00AB CA6502            JZ      EXCUTE
00AE FE14              CPI     B1
00B0 CA7A02            JZ      BRKPT
00B3 FE12              CPI     L1
00B5 CAC908            JZ      USERL
00B8 FE11              CPI     P
00BA CACC08            JZ      USERP
00BD FE17              CPI     R
00BF CAC608            JZ      USERR
00C2 0606       ERROR:  MVI     B,06H   ;IF NOT A COMMAND KEY
00C4 11F808            LXI     D,DSPBFD;  DISPLAY ERROR MESSAGE
00C7 21AB03            LXI     H,ERRMAG
00CA 7E         ERROR1: MOV     A,M     ;TOO MANY BREAK POINTS ENTER
00CB EB                XCHG            ;DISPLAY ERROR MESSAGE
00CC 77                MOV     M,A
00CD EB                XCHG
00CE 23                INX     H
00CF 13                INX     D
00D0 05                DCR     B
00D1 C2CA00            JNZ     ERROR1
00D4 CD5B01     CHECK:  CALL    SCAN    ;WAIT FOR "X" KEY TO ESCAPE
00D7 FE10              CPI     X       ;FROM ERROR
00D9 C2D400            JNZ     CHECK
00DC C38A00            JMP     COMMD
                       ;
                       ;MOVE DATA ROUTINE
                       ;
00DF 0606       MOVE1:  MVI     B,06H   ;MOVE PROMPT TO DISPLAY
00E1 11F808            LXI     D,DSPBFD; BUFFER
00E4 21A503            LXI     H,SINEIN
00E7 7E         MOVE:   MOV     A,M     ;WILL MOVE DATA FROM ONE
00E8 EB                XCHG            ;AREA POINTED TO BY H&L
00E9 77                MOV     M,A     ;TO AN OTHER POINTED TO BY
00EA EB                XCHG            ;D&E. # BYTES TO MOVE IS
00EB 23                INX     H       ;IN REG. B
00EC 13                INX     D
00ED 05                DCR     B
00EE C2E700            JNZ     MOVE
00F1 C9                RET
                       ;
                       ;READ / WRITE ROUTINE
                       ;
00F2 CDD902     WRITE:  CALL    ENTER   ;DATA ENTERED VIA KEYPAD
00F5 CDC901            CALL    INPUT4  ;IS PLACED IN MEMORY AFTER
00F8 CD4401     NEXT:   CALL    DSDATA  ;"X" KEY IS PRESSED AND
00FB CDE001            CALL    INPUT2  ;MEMORY POINTER IS INCREMENTED
00FE DA1101            JC      REDO    ;ONE LOCATION.
0101 32EA08     SECENT: STA     DIGIT2
0104 CDE001            CALL    INPUT2
0107 D20101            JNC     SECENT
010A 3AEA08            LDA     DIGIT2
```